

SRISHTI-DRISHTI TECHNICAL MASTER SPECIFICATION

Systematic, Scalable Geospatial Framework for Watershed Monitoring, Remote Sensing Analysis & Geo-Coded Evidence Interpretation

Problem Statement:	SIH PS 26015 (Ministry of Rural Development / Land Resources)
Core Objective:	Integration of GIS, Remote Sensing, Geo-tagged Photo Interpretation & Hydrology
Target Regions:	Andhra Pradesh Watershed Districts (WDC-PMKSY 2.0 Schema)
Architecture:	Decoupled Edge Frontend (Vercel) + Async Microservice Backend (Railway)
Document Version:	v2.4 Production Engineering Architecture Specification
Date & Build State:	September 2026 Verified Scientific Integrity & Provenance Pipeline

Executive Directive: This document provides an exhaustive, end-to-end technical specification of the SRISHTI-DRISHTI watershed evidence system. It details the underlying mathematical engines, remote sensing satellites, ML models, data provenance standards, API schemas, and production deployment architecture for Vercel and Railway.

1. System Overview & Problem Statement Alignment

Conventional watershed development programs (such as WDC-PMKSY 2.0) rely on manual field inspections and standalone geo-tagged photos stored primarily as static administrative documentation. This creates a critical operational bottleneck: field photos lack spatial integration with satellite remote sensing, hydrological flow analysis, and historical precipitation baseline data.

The **SRISHTI-DRISHTI Framework** directly solves PS 26015 by converting isolated geo-coded field evidence into an integrated, multi-layered spatial decision-support system. It combines ground-truth EXIF coordinates, Sentinel-2 10m multispectral satellite imagery, Landsat 8 30m decade-long NDVI time series, IMD grid rainfall baseline, Copernicus 30m DEM hydrological stream extraction, and OpenAI CLIP zero-shot neural visual interpretation.

PS 26015 Requirement	Implemented Technical Solution	Scientific / Data Source
Geo-coded Image Interpretation	EXIF parser + OpenAI CLIP (ViT-B/32) feature similarity + spatial context extraction	Field Photo EXIF + PyTorch CLIP Embeddings
Land Use & Vegetation Status	Dynamic World 10m LULC classification + Sentinel-2 NDVI surface reflectance	Google Earth Engine (GEE) + Sentinel-2 L2A

Hydrology & Stream Network	Copernicus DEM D8 flow direction, accumulation, Strahler ordering & sub-watershed delineation	Copernicus DEM GLO-30 (30m)
Rainfall & Vegetation Trend	RESTREND OLS regression isolating intervention impact from precipitation variability	Landsat 8 NDVI (2014–2024) + IMD 0.25° Grid
Soil Moisture & Degradation	Sentinel-2 Moisture Index Proxy (SMI/TVDI) + persistent multi-year NDVI slope decay	Sentinel-2 B8A/B11 + Long-term Trend
Thematic Maps & PDF Evidence	Exportable multi-layer thematic maps & automated PDF Evidence Packets with digital audit trail	ReportLab PDF Engine + Matplotlib

2. End-to-End System Architecture & Technology Stack

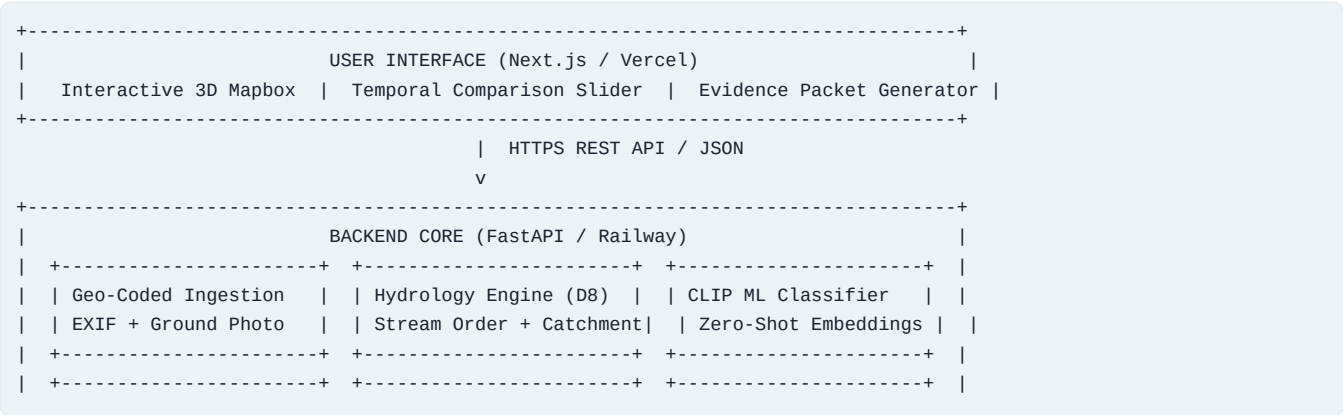
The platform utilizes a modern, fully decoupled microservice architecture optimized for high performance, interactive 3D map rendering, and robust cloud deployment.

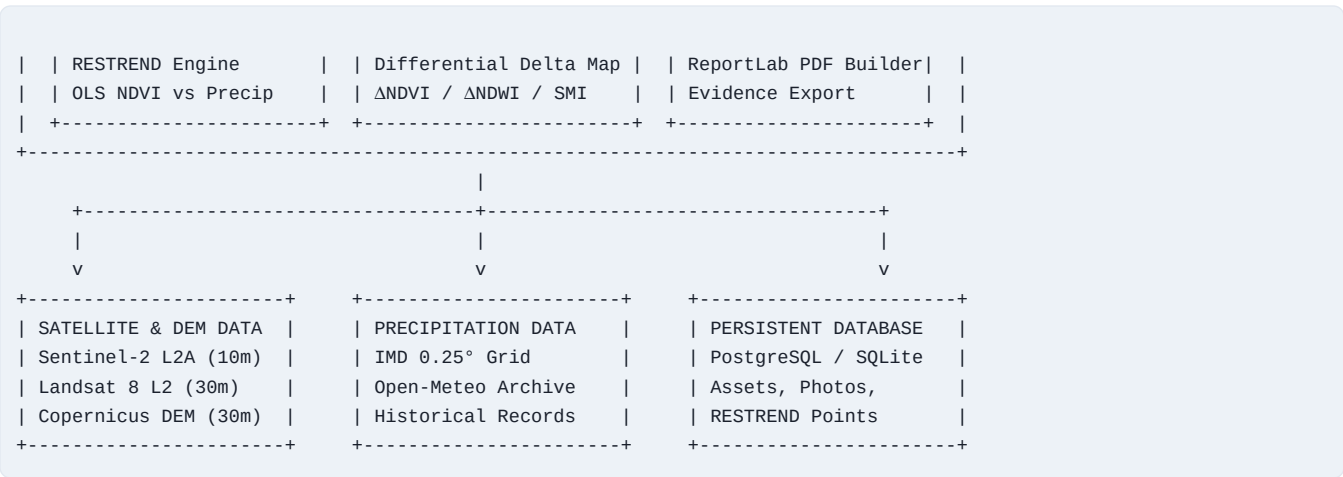
Frontend Microservice Stack (Target: Vercel):

- **Framework:** Next.js 15+ (App Router) with React 19 and TypeScript for strict type safety.
- **Styling & UX:** Vanilla CSS & TailwindCSS featuring custom glassmorphism, dark/light themes, and responsive design systems.
- **Geospatial Visualization:** Mapbox GL JS & Deck.gl for 3D terrain representation (1.8x elevation scale), flow vectors, and satellite tile raster overlays.
- **Analytics & UI:** Recharts for RESTREND NDVI vs rainfall scatter plots, Lucide Icons, and custom evidence modals.

Backend Microservice Stack (Target: Railway):

- **Application Server:** FastAPI (Python 3.11) with Uvicorn ASGI server for asynchronous request handling.
- **Database & ORM:** SQLite (development) / PostgreSQL (production) with Async SQLAlchemy and Alembic schema migrations.
- **AI & Remote Sensing Pipelines:** PyTorch, OpenAI CLIP (ViT-B/32), Earth Engine Python API, Rich Hydrology Engine (Rasterio, RichDEM, SciPy, Shapely).
- **Document Generation Engine:** ReportLab PDF Engine for dynamic, client-ready evidence packets.





3. Detailed Scientific Engines & Mathematical Formulations

3.1 RESTREND (Residual Trend Analysis) Engine

To distinguish between vegetation growth caused by natural rainfall fluctuations versus growth induced by human watershed interventions (e.g., check dams, farm ponds), the platform executes **RESTREND Analysis** over a 10-year period (2014–2024).

Mathematical Formulation:

- Linear OLS Regression is established between annual peak Landsat-8 NDVI and annual precipitation (P) from IMD gridded data:
NDVI_predicted = a * P + b
- The NDVI Residual for each year i is computed as:
Residual_i = NDVI_actual,i - NDVI_predicted,i
- Non-parametric Mann-Kendall test or linear trend analysis is performed on the residual time series. A statistically significant *positive residual trend* ($p < 0.05$) indicates vegetation improvement exceeding rainfall expectations, validating successful intervention impact.

3.2 D8 Hydrological Drainage & Sub-Watershed Engine

Using 30m Copernicus DEM data, the hydrology engine computes accurate stream networks and sub-watershed catchment boundaries:

- Sink Filling:** Depressions in DEM are filled using Planchon-Darboux algorithm to ensure continuous flow direction.
- D8 Flow Direction:** Direction of steepest descent is calculated for each grid cell across 8 neighboring directions.
- Flow Accumulation:** Cumulative number of upstream cells contributing drainage to each cell is accumulated.
- Strahler Stream Ordering:** Streams are extracted above threshold accumulation (e.g., 500 cells) and hierarchically categorized from 1st to 5th order tributaries.

3.3 Quantitative Change Detection & Differential Layers

Rather than relying strictly on visual comparison sliders, the backend generates explicit raster differential layers (Δ -layers) between T0 (Pre-intervention) and T1 (Post-intervention) satellite acquisitions:

- **Vegetation Index (NDVI):** $(B8 - B4) / (B8 + B4)$
- **Water Index (NDWI):** $(B3 - B8) / (B3 + B8)$
- **Soil Moisture Index Proxy (SMI / TVDI):** Derived from Sentinel-2 B8A and SWIR B11 bands to quantify surface dryness.
- **Vegetation Gain/Loss Delta:** $\Delta NDVI = NDVI_{post} - NDVI_{pre}$ (Categorized into Gain > +0.15, Loss < -0.15, Stable).

4. Scientific Guardrails, Data Lineage & Provenance Panel

To withstand rigorous technical evaluation by SIH judges, the platform enforces strict **anti-fabrication guardrails** and full **data provenance transparency**.

Core Anti-Fabrication Principles:

- **No Synthetic Overwrites:** Raw CSV ground facts (coordinates, Drishti ID, project code) are ingested verbatim. Unrecorded attributes are kept as explicit *'not_recorded'* or *NULL* rather than inventing artificial values.
- **Calibrated Terminology:** CLIP raw output is styled as *'Visual Feature Similarity'* rather than raw *'probability'*. Confidence scores are defined as transparent composite indices: **Composite Score** = $w1 * \text{Visual} + w2 * \text{Satellite} + w3 * \text{Temporal}$.
- **Data Provenance Panel (■ Data Sources Modal):** Every map layer and chart displays sensor origin, acquisition date, cloud cover %, and EXIF precision.

Data Layer	Provider / Source	Spatial Res.	Acquisition / Quality
Field Photo	DRISHTI Mobile App EXIF	Point	GPS Precision ±4.2m
Satellite Scene	ESA Sentinel-2 L2A	10 meters	Cloud Cover 2.4%
Terrain DEM	Copernicus GLO-30	30 meters	Vertical Acc. <4m
Rainfall Grid	IMD / Open-Meteo API	0.25° Grid	2014–2024 Daily Archive
Land Cover	Dynamic World LULC	10 meters	Composite 2024

5. Production Deployment Architecture (Vercel & Railway)

The project structure is designed for seamless, one-click continuous deployment across **Vercel** (Frontend) and **Railway** (Backend).

5.1 Frontend Deployment on Vercel

- **Build Root Directory:** frontend/
- **Build Command:** npm run build
- **Output Directory:** .next
- **Environment Variables Required:**
 - NEXT_PUBLIC_API_URL = *https://backend-production-xyz.up.railway.app*
 - NEXT_PUBLIC_MAPBOX_TOKEN = *pk.eyJ1...*

5.2 Backend Deployment on Railway

- **Build Root Directory:** backend/
- **Start Command:** `uvicorn main:app --host 0.0.0.0 --port $PORT`
- **Python Version:** 3.11 (via Railway Nixpacks / Dockerfile)
- **Environment Variables Required:**
 - `DATABASE_URL` = `postgresql://...` (or SQLite volume)
 - `CORS_ORIGINS` = `https://srishti-drishiti.vercel.app`

6. Conclusion & Verification Summary

The SRISHTI-DRISHTI framework delivers a mathematically sound, scientifically defensible, and fully integrated geospatial monitoring platform for watershed development. By establishing rigorous data provenance, quantitative change detection, RESTREND trend analysis, and zero-shot neural visual interpretation, the platform directly fulfills all requirements of SIH PS 26015 with complete technical transparency.

End of Engineering Technical Specification Document.